Syllabus.

Course Schedule:

	Topics/Daily Activities	Readings	Deliverables
Week 1	Modeling with Optimization and basic concepts	**Chapter 1 (Cornuejols)**	
Week 2	Linear Algebra review, unconstrained Opti, local/global optimality, optimality conditions	**Chapter 2 (Winston)**	
Week 3	Root finding methods, algorithms for unconstrained nonlinear optimization (descent methods, coordinate descent, Newton)	**Chapters 12.4, 12.5 (Winston)** **Chapter 5 (Cornuejols)**	**Homework Assignment 1**
Week 4	Convexity: properties and examples: regression, logistic regression, portfolio selection, and Support vector machines, LASSO	**Chapter 12.3 (Winston)** **Appendix A (Cornuejols)**	
Week 5	Constrained nonlinear optimization Duality for constrained nonlinear	**Chapter 12.8 (Winston)**	**Homework Assignment 2**
Week 6	Duality: weak duality, strong duality, KKT conditions	**Chapter 12.9 (Winston)** **Chapter 7 (Cornuejols)**	
Week 7	Algorithms for Constrained nonlinear optimization: Basic descent, projected GD, Penalty methods	**Chapter 12.12 (Winston)**	**Homework Assignment 3**
Week 8	Quadratic Programming, Linear Programing: Examples and modeling, transforming to LP, standard LP	**Chapter 3 (Winston)** **Chapter 2 (Cornuejols)**	
Week 9	**Midterm Exam 3/12**		**Midterm Exam**
Week 10	Geometric interpretation, Background on Simplex	**Chapter 4 (Winston)** **Chapter 2 (Cornuejols)**	
Week 11	Simplex Algorithm, two Phase Procedure, and modeling language (AMPL)	**Chapter 4 (Winston)** **Chapter 2 (Cornuejols)**	**Homework Assignment 4**
Week 12	Duality, simplex method in the dual problem, optimality conditions for LP	**Chapter 6 (Winston)** **Chapter 2.2 (Cornuejols)**	
Week 13	Examples on the use of duality in LP. Integer programming: Modeling and examples	**Chapter 9 (Winston)** **Chapters 11-12 (Cornuejols)**	**Homework Assignment 5**
Week 14	IP: Modeling and examples (Cont'd) Algorithms for IP	**Chapter 9 (Winston)** **Chapters 11-12 (Cornuejols)**	
Week 15	Algorithms for IP	**Chapter 9 (Winston)** **Chapters 11-12 (Cornuejols)**	**Homework Assignment 6**
FINAL		Refer to the final exam schedule in the USC *Schedule of Classes* at classes.usc.edu.	

Goal = Minimum!

week 3.
week 4.

① Gradient 一階 $\nabla f(x)=0$.
確定是 min/max/saddle point

② Newton 二階 $\nabla^2 f(x)$ 判斷.

③ Convexity?
任何 local min = Global min.

無限制 (unconstrained)

現實條件中
大多都是有限制條件的 (constrained)

week 5.
week 6.

① Lagrange Multiplier.
最佳点 = 目標函数的梯度
= constraints 梯度線性組合.
(想往下降, 但 constraints 把你擋住了.)

② Weak / Strong Duality.

③ KKT

Nonlinear Optimisation ≈ quardratic (?)

非線性

- Gradient 梯度 = 最陡的上升方向.

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f(x)}{\partial x_1} \\ \frac{\partial f(x)}{\partial x_w} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{pmatrix}.$$

* 向量.
* 各坐標的函数变化.

- directional derivative 方向导数 =
在 x 上, 沿方向 d 的瞬時变化率.

$$f'(x;d) = \lim_{\lambda \to 0^+} \frac{f(x+\lambda d) - f(x)}{\lambda}.$$

$$f'(x;d) = \nabla f(x)^\top d.$$

內積 = 投影在 d 的变化率.

每個坐標上函数的變化.

- 如何找 Min? 找 $\nabla f(x) = 0$ 的点.

STEP 1 對 f(x) 做微分 → $\nabla f(x) \approx f'(x)$

$$\begin{cases} f'(x) > 0 \\ f'(x) = 0 \\ f'(x) < 0 \end{cases}$$

stationary point 駐点 (曲線斜率 0)
只能確定是極值.

STEP 2 對 f'(x) 做微分 → f''(x) (Hessian 見下方).

判斷是最大/最小/鞍点.

- 找 Stationary Point. 就 d^k, λ.

① Closed Form Solution 封閉解 不用找 d^k, λ.

用 $\nabla f(x) = 0$ 解.

② Descent Method. 下降法. → 找 d^k.

函數有絕對值 → 非光滑.

Prototype descent method

1. Iteration number $k = 0$, initial solution x^0
2. If $\nabla f(x^k) \approx 0$ then stop and **return** x^k
3. Find a descent direction $d^k \in \mathbb{R}^n$
4. Find a step size $\lambda^k > 0$ such that $f(x^k + \lambda^k d^k) < f(x^k)$
5. Set $x^{k+1} \leftarrow x^k + \lambda^k d^k$ and $k \leftarrow k + 1$. **Go to** 2.

$\nabla f(x)$ = 方向 = d^k?

③ Line Search 線搜尋. → 找 λ

- Binary Search

有 upper bound (a), lower bound (b) → $f(a) \cdot f(b) \leq 0$.

$c = \frac{a+b}{2}$ { $f(c) \leq 0$ let $a \leftarrow c$ 用 c 取代 a
$f(c) \geq 0$ let $b \leftarrow c$ 用 c 取代 b

不斷對半縮小區間 直到寬度小於容許誤差 ε

or

$f(c_k) \approx 0$ 也可以?

- Newton Method.

- 解一階可微函數 $\Longleftrightarrow$ 解 $g(x)=0$

用切線和x軸交點當作更好的解

推導

STEP 1 $g(x) \approx g(x_k)+g'(x_k)(x-x_k)$ (在x_k的切線)

STEP 2 $g(x_k)+g'(x_k)(x-x_k)=0$

STEP 3 $x-x_k=-\frac{g(x_k)}{g'(x_k)}$

一階Taylor展開

$f(x) \approx f(x_0)+f'(x_0)(x-x_0)$

$\approx$ 起點高度 + 在x_0的斜率 × 走了多遠

$$x_{k+1}=x_k-\frac{g(x_k)}{g'(x_k)}$$

算出 $-\frac{g(x_k)}{g'(x_k)}$ 代回算出 x_{k+1}

Repeat 直到 $-\frac{g(x_k)}{g'(x_k)}<\varepsilon$

- 解一階可微函數 $\Longleftrightarrow$ 解 min/max 問題.

$\Longleftrightarrow$ 解 $g'(x)=0$

taylor 一階展開.

$$f(x_k+\Delta x) \approx f(x_k)+f'(x_k)\Delta x+\frac{1}{2}f''(x_k)(\Delta x)^2$$

(一階展開)

$$x_{k+1}=x_k-\frac{g'(x_k)}{g''(x_k)}$$ (算法同理如上)

$$=x_k-H^{-1}(x_k)\nabla g(x_k)$$

展開階數	用什麼近似	幾何意思
0 階	常數	一個點
1 階	直線	切線
2 階	拋物線	曲率
更高階	多項式	更精準

補 Newtod Method 是 taylor 展開式的應用.

Binary Search	Newton Method.
⊕ Assume $f(x)$ exist and is continuous.	⊖ Assume $f'(x)$, $f''(x)$ exist
⊕ Always works	⊖ need additional assumption e.g. $f''(x) < M$
⊕ Initial iteration → long programs	⊖ Initial Iteration → no idea
⊖ high precision? → slow / constant effort	⊕ → fast coverage (if close to minimum point)

- steepest descent 最速下降法.

在每一步選擇時的下降最快的方向 $\Leftrightarrow -\nabla f(x)$

再配合 line search 找入.

$$d_k = -\nabla f(x^k)$$

- Hessian 多變數函數的二階導數總表(矩陣) *未學.

紀錄每個(混合)方向的彎曲程度

{ Gradient $\nabla f(x)$ → 看曲線的斜率. $f'(x)$

Hessian $\nabla^2 f(x)$ → 看曲線的彎曲程度. $f''(x)$

$$d_k = -\nabla^2 f(x^k)^{-1} \nabla f(x^k) \longrightarrow \frac{g'(x)}{g''(x)}$$

(一維) $\nabla^2 f(x)$ { >0 min

$=0$ stationary point

<0 max.

(多維) $H = \nabla^2 f(x)$ (矩陣)

任意方向. $d^T H d$ {

>0 Positive Definite 正定 local min

像碗一樣每一方向都向上.

≥ 0 Positive Semi-Definite 半正定

有方向是平的. 要看更高階項

<0 Negative Definite local max.

Indefinite 不定 saddle point

有向上也有向下的

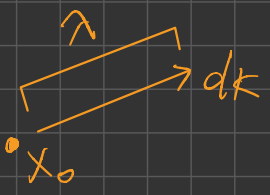
Steepest Descent	Newton Methods.
⊕ 1st order $\nabla f(x)$	⊖ 2nd order $\nabla f(x)$ $\nabla^2 f(x)$
⊕ always converge	⊖ may not converge.
⊕ cheap iteration	⊖ expensive iteration $\nabla^2 f(x)^{-1}$
⊕ initially = great progress	⊖ no guarantee (unless additional assumption)
⊖ slow = high quality solution	⊕ once near, is fasten

Target: Find min $f(x)$ (多維情況下沒辦法用兩次微分解).

使用 Descent Methods.

- 隨机選取 x_0.
- 計算 $\nabla f(x)$
- 選方向 d_k. → 用 descent methods, $d_k = -\nabla f(x_k)$

  Hessian 的 Newton, $d_k = -\nabla^2 f(x_k)^{-1} \nabla f(x_k)$
- 選 λ (在 d 方向上走多遠) → 用 line search. { Binary Search / Newton Method.
- 更新 $x_{k+1} = x_k + d_k + \lambda_k$. (一直往極值的方向走).
- 重複直到 $\nabla f(x) = 0$. (已知是極值只是不知道是大是小).
- 用 Hessian 判斷 (確定是最大/最小/鞍点) How?!



How to identify good solution?

Global Min $f(x) \le f(y)$

⇒ find stationary point - Locally, fun is flat.

$f(x) = 0$ → flat #

Duality

{ 原問題：直接想解的問題
  對偶問題：從限制條件的角度看問題

原問題 ⟷ 對偶問題

$\max c^T x$ s.t. $Ax \le b$, $x \ge 0$

$\min b^T y$ s.t. $A^T y \ge 0$, $y \ge 0$

- Weak Duality　dual objective ≤ primal objective

對任何可行的 x 和 y，
對偶問題的目標函數值
是原問題的上界/下界。

原	對偶
max ⟶	上界 (min的不得欠)
min ⟶	下界 (max的相欠)

- Strong Duality　primal optimum = dual optimum

如果兩邊都有最佳解

$\max c^T x = \min b^T y$.

Dual 解法流程

① 写出primal（通常是题目提供）

② Compute Dual

③ 找出optimal condition

- 列出 PF (primal function) / DF (dual function)
- 列出 CS
- 看哪几组能使两式可解

extra 解完代回 DF. 有一项 x → 未必是 optimal.

weak duality.

如何确认是 optimal value?

- 原始可行性 PF
  代值代回 PF constraints 都符合.
- 对偶可行性 DF
  代值代回 DF constraints 都符合.
- 互补松弛性
  代值代回 CS 满足:
  对每个 y_i (primal slack i) $=0$
  满足 { slack >0 则 $y=0$
  $y>0$ 则 primal constraint $i=0$.

) weak duality.

- object value 相等) strong duality.

$\min c^\top x$	$\max y^\top b$
✓ $a_i^\top x \geq b_i$	$y_i \geq 0$
$a_i^\top x \leq b_i$	$y_i \leq 0$
$a_i^\top x = b_i$	y_i free
$x_j \geq 0$	$y^\top A_j \leq c_j$ ✓
$x_j \leq 0$	$y^\top A_j \geq c_j$
x_j free	$y^\top A_j = c_j$

補 觀念釐清.

① LP / QP

QP 與 LP 的解題差異

特性	線性規劃 (LP)	二次優化 (QP)
目標函數	一次式，例如 $6x_1 + 5x_2$	二次式，例如 $(x_1 + x_2)^2$
圖形長相	像一塊平坦的斜坡	像一個碗或拋物面
最優解位置	一定在可行區域的頂點 (Corner)	可能在區域內部（碗底）或邊界
核心工具	對偶法、單體法 (Simplex)	梯度 ($\nabla f = 0$)、牛頓法

② 開放變數的數量 (x_i, $i = n$)

{ LP = 使用 dual. Simplex.

QP = 使用 Hessian 解題.

Simplex Method 單形法

- Basic Solution

問題 min/max _____

$s.t.\ Ax = b,\ x \geq 0$

$(A_{m\times n},\ b_{m\times 1})$

$\begin{cases} n & \text{variables} \\ m & \text{constraints} \end{cases} \rightarrow$ assume $n > m$ (代表有無限多解)

Def 1 把 x 拆成 $\begin{pmatrix} X_B \\ X_N \end{pmatrix}$ 基本變數 / 非基本變數

強制 $X_N = 0$, $X_B = B^{-1}b$

確認是最佳解的條件.

Def 2 所有 $X_B \geq 0$, 那可行. (P) 滿足

Def 3 有最佳解的話, 必是 extreme point.

- Pivoting 樞紐運算

從目前的 extreme point 移動到下一個頂點.

將一個 X_B 變成 X_N (=0), 一個 X_N 變成 X_B.

Reduce Cost > 0

//

- Adjacent (相鄰): 每次只換一個 X_B 和一個 X_N
- Better (更好): 選擇 Reduced cost 是 (+) 的變數
- Feasible (可行): 用 Ratio Test 決定哪一變數會被刪除.

- Duality

從CS來看：$X_N = 0$ 代入 x_i · 任何東西 $= 0$

$X_B \geq 0$ x_i (　　) $= 0$

這一部份須 $= 0$

- Reduce Cost

剩下DF還沒滿足 $\begin{cases} X_B = y^T B - C^T B = 0 \\ X_N = y^T N - C^T N \geq 0 \text{ ? 檢查!} \end{cases}$

- Initialisation

用於一開始找不到符合pf的解.

$Ax = b$ 來看 $\begin{cases} b \geq 0 = x = 0, \text{ CS變數變成 } b. \\ b < 0 = \text{Two Phase Method.} \end{cases}$

- Phase 1

加入 t 來解. 如有解且 $t \neq 0$, 則 infeasible.

- Phase 2

" 如有解且 $t = 0$, 則用算法

Branch and Bound 分支界定法

① Optimisation problem 最佳化
② Combinatorial problem 組合最佳化
解是離散的 ex: subsets, permutation
③ Mixed-Integer problem 混合整数.
变数必须是整数 (0,1) (部份)

解決現實中很多只能有整数的問題
{ 最佳化問題 classification

Mixed-Integer Linear Program

→ NP-HARD. LP Solvable in polynomial

MILP

拿掉 整数限制 ↓

Relaxed

Mixed-integer linear programming problem (MILP)

$$\min_{x,z} \boldsymbol{a}^\top \boldsymbol{x} + \boldsymbol{c}^\top \boldsymbol{z}$$
$$\text{s.t. } \boldsymbol{A}\boldsymbol{x} + \boldsymbol{G}\boldsymbol{z} \le \boldsymbol{b}$$
$$\boldsymbol{x} \in \mathbb{R}^n,\ \boldsymbol{z} \in \mathbb{Z} \text{ 整数}$$

operation $\approx 2^m$

Linear programming (LP) relaxation

$$\min_{x,z} \boldsymbol{a}^\top \boldsymbol{x} + \boldsymbol{c}^\top \boldsymbol{z}$$
$$\text{s.t. } \boldsymbol{A}\boldsymbol{x} + \boldsymbol{G}\boldsymbol{z} \le \boldsymbol{b}$$
$$\boldsymbol{x} \in \mathbb{R}^n,\ \boldsymbol{z} \in \mathbb{R} \text{ 实数}$$

operation $\approx (n+m)^3 \approx n^6$

Preliminaries (求者的關係)

真正最佳值 $z^f(S) = \min\{f(\boldsymbol{x}) : s \in S\}$ (P)
Relaxed後最佳值 $z^g(R) = \min\{g(\boldsymbol{x}) : s \in R\}$ (Q)
with $S \subseteq R$ and $g(\boldsymbol{x}) \le f(\boldsymbol{x})$ for all $\boldsymbol{x} \in S \Leftrightarrow$ (Q) is a relaxation of (P)

Observations

- $z^g(R) \le z^f(S)$ Relaxed 值必比較樂觀.
- If $\bar{\boldsymbol{x}} \in S$, then $f(\bar{\boldsymbol{x}}) \ge z^f(S)$ 整数解 → 最佳解.
- If $\boldsymbol{x}^* \in S$ optimal for Q and $f(\boldsymbol{x}^*) = g(\boldsymbol{x}^*)$, then $\boldsymbol{x}^*$ is optimal for P
- If $S = S_1 \cup S_2$, then $z^f(S) = \min\{z^f(S_1), z^f(S_2)\}$

↳ 把 S 分成 S_1, S_2

Branch and Bound

Definition

$S \subseteq \mathbb{R}^n$ feasible region of problem
A list of subproblems to be solved → 待判清單.
z_{best} best obj value found, x_{best} best sol found → 最好的答案(x)
→ 最好的目標函數值 [x]

① Initialization

問題是Min, z_{best} 越小越好.

Step 0 $A \leftarrow \{S\}$, $z_{\text{best}} = \infty$, $\boldsymbol{x}_{\text{best}} = \emptyset$

原始問題 任何z都會更好. 空集合.

② Selecting & Solving.

Step 1 Select a subproblem T from A
$A \leftarrow A \setminus T$
Solve a relaxation R_T of T

③ Pruning (Bound)

Step 2 If $z(R_T) \geq z_{\text{best}}$, then **go to** 1

→ 直接放棄這個分枝.

④ Upgrading New Ans.

Step 3 If $\boldsymbol{x}^*(R_T) \in S$, then $z_{\text{best}} \leftarrow z(R_T)$, $\boldsymbol{x}_{\text{best}} \leftarrow \boldsymbol{x}^*(R_T)$; **go to** 1

⑤ Branching.

如果算出來 $\notin S$, 但 z_{best} 有比較小. → 代表有潛力!

Step 4 Divide T into subproblems: $T = \bigcup T_k$ → 拆分問題
$A \leftarrow A \cup \{T_1, \ldots, T_k\}$ → 把拆分後的問題丟回A.
go to 1

Branch and cut ???

分枝策略

- How? (目的：縮短求解的時間)
  - ① Strong Branching：實際計算每個分枝的impact.
  - ② Pseudo Costs 偽成本：根據歷史數據估計
- Which? (目的：縮短時間+節省記憶)
  - ① Depth-first：找可行解快.
  - ② Best Bound：減少node數.

Heuristics 啟發式演算法

- 目的：
  - 快速建立上界.
  - 太過複雜時，先專注找出符合所有條件的解.
  - 提升 LP relaxed 的品質

① Rounding

② Neiborhood Search ???

③ Feasibility Pump.

Presolve 預處理

① Remove redundant constraints. 刪除沒用的限制.

② Probing. 固定某變數, 看結果有不有有影響.

③ Conflict Graph 找出不相容的變數

Branch and cut 縮小 LP relaxation 的範圍.
使整數解更容易被找到.

Gomory Cut
利用整數性質. 進行向下取整的無條件捨去.

for general questions.

for specific questions.

Clique Cut
利用圖形結構來排除非整數解的手段

Constrained Optimisation

Lagrangian Relaxation

$$\begin{aligned}
&\min\ f(x) && && \text{(1a)}\\
&\text{s.t. } a_i^\top x \le b_i && i=1,\dots,m && \text{(1b)}\\
&\quad h_i^\top x = d_i && i=1,\dots,p && \text{(1c)}\\
&\quad x \in X. && && \text{(1d)}
\end{aligned}$$

有此約束條件整合進目標函數中. penalties / 原本值之有取捨空間

↳ 本來就會比 f(x) 小.

$$\mathcal{L}(x,\lambda,\mu) = f(x) + \sum_{i=1}^{m} \underbrace{\lambda_i}_{\geq 0}\left(a_i^\top x - b_i\right) + \sum_{i=1}^{p} \underbrace{\mu_i}_{\text{free}}\left(h_i^\top x - d_i\right).$$

Duality

定義 $w(\lambda,\mu)$ 是 $\mathcal{L}$ 的極小值

- Primal Prob = $\min_x w(\lambda,\mu)$ → $\min_x (\max_{\lambda,\mu} \mathcal{L}(x,\lambda,\mu))$
- Dual Prob = $\max_{\lambda,\mu} w(\lambda,\mu)$ → $\max_{\lambda,\mu} (\min_x \mathcal{L}(x,\lambda,\mu))$

(λ,μ) 的極小值是原問題最優解的 lower Bound (min)

- weak dual : $w^* \le z^*$
- strong dual : $w^* = z^*$ (if f is convex)

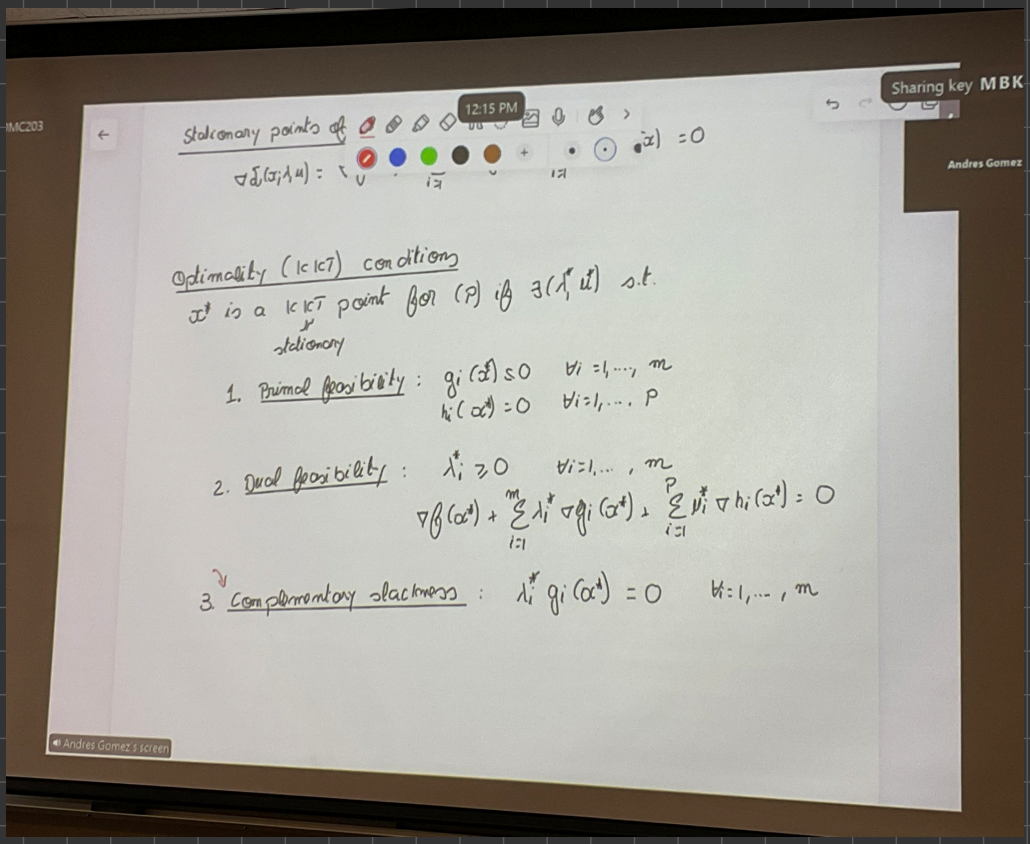
Duality Gap = Primal value − Dual value

Optimal Condition

- 在强对偶成立时，最优解 x^* 是 $w(\lambda, \mu)$ 的平稳点。

$$\nabla f(x) = 0$$

- 最优解满足 P.F. D.F. C.S.



△ 如果 x^* 是 optimal value → 是 KKT point.

KKT point 不代表 optimal

Methods of Feasible Directions. 可行方向法.

Prototype descent method for constrained minimization

1. Iteration number $k = 0$, initial feasible solution x^0 such that $Ax^0 \leq b$.
2. If x^k is a KKT point, then stop and **return** x^k.
3. Find a feasible descent direction $d^k \in \mathbb{R}^n$
4. Find a feasible step size $\lambda^k > 0$ such that $f(x^k + \lambda^k d^k) < f(x^k)$
5. Set $x^{k+1} \leftarrow x^k + \lambda^k d^k$ and $k \leftarrow k + 1$. **Go to** 2.

In step 3-4, we need to find a direction d^k and λ^k such that:

(a) d^k is a descent direction at x^k, i.e.,

$$\nabla f(x^k)^\top d^k < 0.$$

(b) d^k is a feasible direction, this is,

$$A\left(x^k + \epsilon d^k\right) \leq b$$

for some $\epsilon > 0$.

d^k 符合.
① Descent $\nabla f(x^k) d^k < 0$
② Feasible $x^k + d^k$ 在 $Ax \leq b$ 裡.

λ^k 符合.
① $f(x^k + d^k) < f(x^k)$
② 確保不出可行區域.
解 $\min f(x^k + \lambda d^k)$, $0 \leq \lambda \leq 1$

Interior Point Methods 內點法.

加上 Logarithmic Barrier Function

x 靠近邊界時, $g(x) \approx 0$, $\ln \approx -\infty$, 取值 $\approx \infty$

$$\min \; f(x) - \sum_{i=1}^{m} \frac{1}{t} \ln(-g_i(x)). \qquad \text{(Barrier)}$$

決定非負力有多強.

t { 小 逐漸逼近 t. Newton Method 求解
大 近似效果好, 但很難解

總結對比

特性	可行方向法	內點法 (Interior Point)
路徑	沿著可行域的邊界或內部移動	始終從可行域的中心穿過
終止點	隨時停止，結果皆為可行解	疊代未完成前，解可能不夠精確
適用場景	線性約束、需要隨時有可行解時	大型、複雜非線性約束問題

5/30 Final Recap.

問題本質 $\min f(x)$

$s.t.\ g(x) \le 0 \quad i = 1 \dots m$

$x_j \in \mathbb{Z} \quad j = 1 \dots p.$

圖=主解. Definition補充

h 定的公式:

代表input/output的關係

θ h函數的parameter

輸入的feature. response 目標值

Training ML model $\{a_i, y_i\}_{i=1}^{m}$

推測 conjecture $h = \mathbb{R}^n \to \mathbb{R}$ ← 輸入n維實數. 輸出實數(預測值)

↳ 待驗證. $h_\theta(a_i) \approx r$ ← 現階段生猜法 通常 $r = y_i$

θ parameters (e.g. regression coefficients)

訓練目標 Training $\min \sum_{i=1}^{m} L(h_\theta a_i - y_i)$ ← 找到一組θ使預測-真實的L最小

$s.t.\ \theta \in S$ ← 屬於一個大集合.

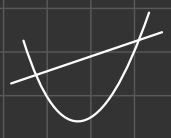
↑ 定義好函數.

How to identify optimal solution?

$\min f(x)$

$s.t.\ x \in \mathbb{R}^n$

a. convexity



$f''(x) \ge 0 \quad \forall x$

- 1D
- 2D+
  - $h(x) = a^T x + b$ are convex.
  - $h(x) = f(a^T x + b)$ is convex. if f is convex. e.g. $h(x) = (a^T x + b)^2$, $\ln(1 + e^{-a^T x + b})$
  - $h(x) = \sum_{i=1}^{m} \lambda_i f_i(x)$ is convex if all $\lambda_i \ge 0$ & all f_i are convex.

常見=維以上凸函數組合.

b. Stationary point $\nabla f(x) = 0$

if f is convex, then x^* minimize f is $\nabla f(x^*) = 0$

Algorithm for unconstrained optimization

why 有的条件不有 一个同都有?

① Pick x^0

for every iteration

Pick d such that $d^T \nabla f(x^k) < 0$

Pick step size α

$x^{k+1} = x^k + \alpha d$.

d1. Steepest descent → extremely fast.

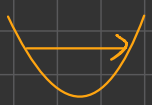
$d^1 = -\nabla f(x^k)$

$d^2 = -\nabla^2 f(x^k)^{-1} \nabla f(x^k)$

d2 Newton Methods.

$\alpha = \varepsilon$ → ?

α optimal for $\min_{\alpha \geq 0} f(x^k + \alpha d)$



↑ unconstrained.

↓ constrained.

Linear Programming.

$$(P)\quad \min_x c^T x \quad s.t.\ Ax \geq b \qquad \Longleftrightarrow \qquad (D)\quad \max y^T b \quad s.t.\ y^T A = c^T,\ y \geq 0$$

Optimal Conditions

x is optimal for (P) if and only if $\exists y$

- PF: $Ax \geq b$
- DF: $y^T A = c^T$ $y \geq 0$
- CS: $y_i(a_i^T - b_i) = 0$ (equiv $= y^T b = c^T x$)

Simplex method.

① guess solution x: $Ax \geq b$. (PF)

② Use CS to get y $y_i(a_i^T x - b_i) = 0$ (CS)

③ Verify $y^T A = c^T$ (DF)

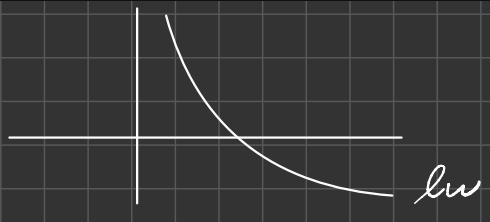
$y \geq 0$ [Yes = optimal ☑ / No = $y_i < 0$, update $x \to a_i^T x = b_i$

演算法類型	適用問題類型	運作邏輯與特色
Simplex Method (單體法)	有約束 線性規劃 (LP)	沿著可行域的**邊界頂點（Vertices）**一個一個跳轉，直到找到最優點。
Interior Point Method (內點法)	有約束 線性/非線性規劃	從可行域的內部直接穿過去，利用剛才提到的對數障礙函數限制自己不超出邊界。
Newton's Method / Gradient Descent	無約束 (Unconstrained)	沒有邊界限制，單純看目標函數的斜率和彎曲度來尋找最低點。

Interior Point

$$\min c^T x - \frac{1}{t}\sum_{i=1}^{m} \ln(a_i^T x - b_i)$$

solve and increase t



Lagrangian

原問題 $\min f(x)$

$s.t.\ g(x) \le 0,\ \lambda \ge 0.$

L 函數 $L(x;\lambda) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x)$

$z^*(\lambda) = \min_x L(x;\lambda)$ 對於固定的 λ，求 L 中最小的 x.

- weak dual : For any x satisfying $g_i(x) \le 0$ and $\lambda \ge 0$ (PF)

$$f(x) \ge f(x) + \sum_{i=1}^{m} \lambda_i \underbrace{g_i(x)}_{\le 0}$$

$$\boxed{z^*} = \min f(x) \quad s.t.\ g_i(x) \le 0 \quad \ge \quad \boxed{z^*(\lambda)} = \min f(x) + \sum \lambda_i g_i(x)$$

- strong dual : $\bar{x}(\lambda)$ optimal

equal if $\bar{x}(\lambda)$ satisfies $\lambda_i g_i(\bar{x}(\lambda)) = 0$.

Optimal Condition

if f is convex, g_i are convex and regularity condition

then x is optimal if $\exists \lambda$ such that.

- PF $g_i(x) \le 0$
- DF $\lambda \ge 0$ and $\nabla f(x) + \sum \lambda_i \nabla g_i(x) = 0$
- CS $\lambda_i g_i(x) = 0 \quad \forall i = 1, \dots, m.$

Algorithm 1 Enumeration (Not Good) 窮舉法.

For all combination of $\lambda_i = 0$ or $\lambda_i > 0$

solve unconstrained system of equation

Algorithm 2 LP Approximation 線性逼近

$\min f(x)$ s.t. $g_i(x) \le 0$ and iterate x^k $\xrightarrow{\text{Approximate}}$ (LP)

$\min \nabla f(x^k)^T x$ s.t. $\nabla g_i(x^k)^T x \le 0$

↓ solve

x_{LP}.

$$x^{k+1} = x^k + \alpha (x_{LP} - x^k)$$

Algorithm 3 非線性逼近

$\min f(x)$ s.t. $g_i(x) \le 0$ and iterate x^k $\xrightarrow{\text{Approximate}}$ (unconstrained)

$\min f(x) + \sum_{i=1}^{m} h_i(g_i(x))$ e.g. $\ln$ Depend on x^k.

↓ solve

$x_{unconstrained}$.

$$x^{k+1} = x^k + \alpha (x_{unconstrained} - x^k)$$

Mixed - Integer optimization

$\min c^T x$ s.t. $Ax \ge b$

$x_j \in \mathbb{Z}$ $j = 1, \dots, p$, ← 刪掉等.

先當作沒有整數限制, 如有小數限制再來Branch.

Branching.

Step 1 Approximate LP relaxed ~~$x_j \in \mathbb{Z}$~~

Step 2 get relaxation and get x_{LP}

- Infeasible X
- if x_{LP} is integer and good quality ✓
- otherwise pick $(x_{LP})_j \notin \mathbb{Z}$, create two subproblems, where $x_j \le 0$ or $x_j \ge 1$ and repeat